

Automatic Invoice Generator

Team : Guardians

1.Problem Statement

Small businesses often rely on manual billing, which is time-consuming, error-prone, and difficult to manage. This project provides an automated invoice generation system with inventory management, customer record management, and business analytics to improve efficiency and accuracy.

2.Objectives

- To automate the invoice generation process.
- To reduce manual work and billing errors.
- To manage customer and product records efficiently.
- To generate PDF invoices and send them via email.
- To store data securely using JSON.
- To provide customer purchase insights.
- To ensure accurate data through input validation.
- To develop a simple and user-friendly billing system.

3.Features

- **Customer Management:** Manage customer records.
- **Product Management:** Manage products and inventory.
- **Customer Intelligence:** Analyze customer purchase history.
- **Invoice Generation:** Generate invoices with GST calculations.
- **PDF Generation:** Create PDF invoices.
- **Email Sending:** Email invoices to customers.
- **Search:** Search customer, product, and invoice records.
- **Reset:** Clear input fields.

- **Sales Dashboard:** Display total sales, total invoices, top-selling products, and monthly sales summary.
- **JSON Storage:** Store and retrieve data using JSON.
- **Validation:** Validate user inputs.
- **Menu-Driven Integration:** Access all modules through a simple menu.

4. Module Requirements

Module 1: Customer Management

Purpose: Manage customer information.

Requirements:

- Add customer details (ID, name, phone, email, address, GST if applicable).
 - Validate customer information.
 - Save and retrieve customer records from customers.json.
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Module 2: Product Management

Purpose: Manage product inventory.

Requirements:

- Add product details (ID, name, category, price, stock).
 - Validate product information.
 - Save and retrieve product records from products.json.
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Module 3: Customer Purchase Intelligence

Purpose: Analyze customer purchase history.

Requirements:

- Read customer purchase history from invoices.json.

- Display purchase summary, total spending, frequently purchased products, and discount eligibility.
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Module 4: Invoice Generation

Purpose: Create customer invoices.

Requirements:

- Retrieve customer and product details.
 - Check customer purchase history.
 - Calculate subtotal, GST, discounts, and final amount.
 - Generate a unique invoice and save it to invoices.json.
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Module 5: PDF Generation

Purpose: Generate invoice PDF.

Requirements:

- Read invoice details.
 - Create a professional invoice PDF.
 - Include company, customer, product, GST, discount, and total amount details.
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Module 6: Email Sending

Purpose: Send invoice through email.

Requirements:

- Retrieve customer email.
 - Attach the invoice PDF.
 - Send the email and handle errors.
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Module 7: Search Module

Purpose: Search stored records.

Requirements:

- Search customers by ID or name.
 - Search products by ID or name.
 - Search invoices by Invoice ID or Customer ID.
 - Display matching records.
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Module 8: Reset Module

Purpose: Clear entered data.

Requirements:

- Reset current input fields.
 - Clear temporary invoice data.
 - Confirm before resetting data.
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Module 9: Sales Dashboard

Purpose: Display sales statistics and business insights.

Requirements:

- Read invoice data from invoices.json.
 - Display total sales and total invoices.
 - Show top-selling products.
 - Display monthly sales summary.
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Module 10: JSON File Handling

Purpose: Manage project data.

Requirements:

- Create, read, write, and update JSON files.

- Handle missing or invalid files.
 - Provide common file-handling functions for all modules.
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Module 11: Main Menu (Integration)

Purpose: Connect all project modules.

Requirements:

- Display the main menu.
- Navigate to all modules.
- Coordinate data flow between modules.
- Exit the application safely.

5.Non-Functional Requirements

- The system should be simple and user-friendly.
- The application should generate invoices quickly and accurately.
- Data should be stored securely and reliably.
- The system should validate user inputs to minimize errors.
- The application should be easy to maintain and update.
- The system should perform efficiently with multiple records.
- The application should be compatible with standard desktop systems.

6.Product Backlog Grooming

Product Backlog ID	User Story	Priority	Estimated Effort
PB-01	As a user, I want to add, update, search, and validate customer records.	High	3 Hrs

PB-02	As a user, I want to add, update, and validate product details with automatic inventory updates.	High	3 Hrs
PB-03	As a user, I want to analyze customer purchase history for business insights.	Medium	4 Hrs
PB-04	As a user, I want the system to generate invoices with GST and total calculations.	High	4 Hrs
PB-05	As a user, I want to generate invoices in PDF format.	High	3 Hrs
PB-06	As a user, I want to send invoices to customers via email.	Medium	3 Hrs
PB-07	As a user, I want to search customer, product, and invoice records.	Medium	2 Hrs
PB-08	As a user, I want to reset all input fields for new entries.	Low	1 Hrs
PB-09	As a user, I want to view a sales dashboard to analyze business performance.	Medium	3 Hrs
PB-10	As a user, I want to store and retrieve data using JSON.	High	3 Hrs
PB-11	As a user, I want all modules integrated into a menu-driven application.	High	3 Hrs
PB-12	Testing, bug fixing, and final documentation.	High	5 Hrs

7.Sprint Planning

- **Sprint 1: Planning Phase (Duration: Before Submission / 1–2 Days)**
 - Requirement analysis and module finalization
 - System design and architecture preparation

- Preparation of project report (Objectives, Features, Requirements, etc.)
- Final submission of report
- **Sprint 2: Development Phase (Duration: 2-3 Days)**
 - Implementation of Customer Management module
 - Implementation of Product Management module
 - Development of Invoice Generation with GST calculation
 - PDF generation of invoices
 - Email sending functionality
 - JSON-based data storage implementation
 - Search and Reset functionalities
 - Implementation of Sales Dashboard module.
 - Menu-driven system integration
- **Sprint 3: Testing Phase (Duration: 1–2 Days)**
 - Testing of all implemented modules
 - Bug identification and fixing
 - Validation of inputs and outputs
 - Final system integration and verification

8.Workflow

Start



Main Menu Display



Select Option



1. Create Invoice

- Enter Customer ID
- Customer Intelligence
- Add Products
- Calculate Bill
- Generate Invoice
- Save (invoices.json)
- Generate PDF
- Send Email

2. Search Invoice

- Enter ID / Customer ID
- Display Result

3. Sales Dashboard

- Display Total Sales
- Display Total Invoices
- Display Top Products
- Display Monthly Sales

4. Reset

- Clear temporary data

5. Exit

9. Technologies Used

- Python – Core programming language
- JSON – Data storage for customers and products
- FPDF / ReportLab – PDF invoice generation
- SMTP – Email sending
- VS Code – Development environment

10. Acceptance Criteria

- Customer and product data is stored and retrieved correctly.
- Invoice is generated with correct bill calculation.
- PDF invoice is created successfully.
- Email is sent to the customer without errors.
- System handles invalid inputs properly and runs without crashing.
- Sales dashboard displays accurate sales statistics.
- Dashboard updates automatically after new invoices are generated.

11. Future Scope

- Integration of online payment systems (UPI/cards)
- AI-based product recommendations

- Barcode scanning for faster billing
- Real-time stock management with alerts
- Interactive charts and graphs for sales analysis.

12.Task Distribution

